DNS SEC(URITY) HACKATHON

# DNSSECretary

DNSSEC errors, found, explained and fixed before customers notice

**Team Port 53**

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THE PROBLEM

# One small mistake takes a signed domain offline

01 Expired signature

02 Wrong DS record

03 Rollover too fast

04 Broken NSEC3 chain

THE RESULT

# 10–15%*

## of .rs domains are signed

Fear keeps adoption low

* Source: Dijana from RNIDS (Webinar)

MEET DNSSECRETARY

# A web app that keeps signed zones healthy

It watches every zone, explains each error and hands the admin a tested fix.

BUILT FOR

## DNS hostings

Hundreds of signed customer zones in one view.

THE GAP

# Today the customer finds it first

### Today

- The customer calls: “my site is down”
- Tools like DNSViz show what is wrong
- The admin works out why, and the fix

### With DNSSECretary

- The admin sees it within 20 seconds
- Explained, with who has to fix it
- The fix is already tested

THE SOLUTION: A WEB APP FOR HOSTING ADMINS

# Every zone, every 20 seconds

| | | |
|---|---|---|
| 01 | **Detect** | Walk the chain of trust. Ask every name server and a validating resolver. |
| 02 | **Diagnose** | Plain language: what breaks, why, and who has to fix it. |
| 03 | **Fix** | A short shell script the admin can read and edit. |
| 04 | **Verify** | Tested in a sandbox first. One click runs it live, then checks again. |

HOW WE TEST

# Safe to test, easy to show

**30**
broken zones, one per error type

**40**
customer zones in a made-up hosting

**~5 s**
to test a fix in the sandbox

**30/30**
errors detected, each with a verified fix

Our own root, TLD and validating resolver in Docker. No internet, no traffic to real DNS.

TECH STACK

# Under the hood

`scan → check → diagnose → fix script → sandbox → execute → verify`

ENGINE

## Java 21 + dnsjava

Queries, RRSIG checks, DS digests

WEB APP

## Javalin + React

REST and live updates, TypeScript

SIGNING

## BIND tools

dnssec-signzone and dnssec-verify

LAB

## Docker Compose

NSD and Unbound on an internal network

SANDBOX

## One container

No network, every lab server inside

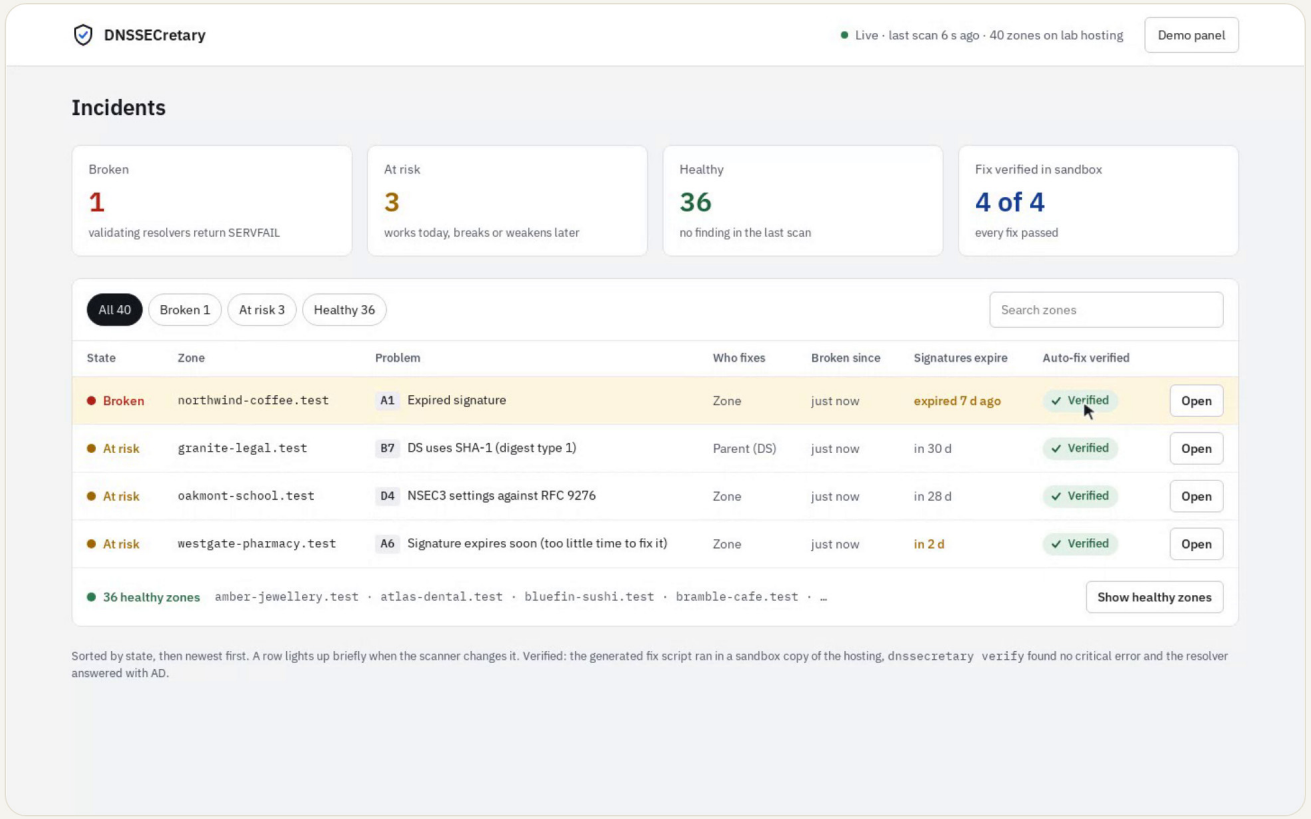
TESTS

## JUnit 5 + CI

The whole lab, from scratch, on every change

LIVE DEMO

# Broken to healthy in 40 seconds




1 Break a customer zone

2 Fix verified in the sandbox

3 Open: diagnosis and script

4 Execute on hosting: healthy

DNSSECRETARY

# DNSSEC protects people only if operators dare to keep it on

DNSSECretary makes it easy to keep on

Thank you · Team Port 53